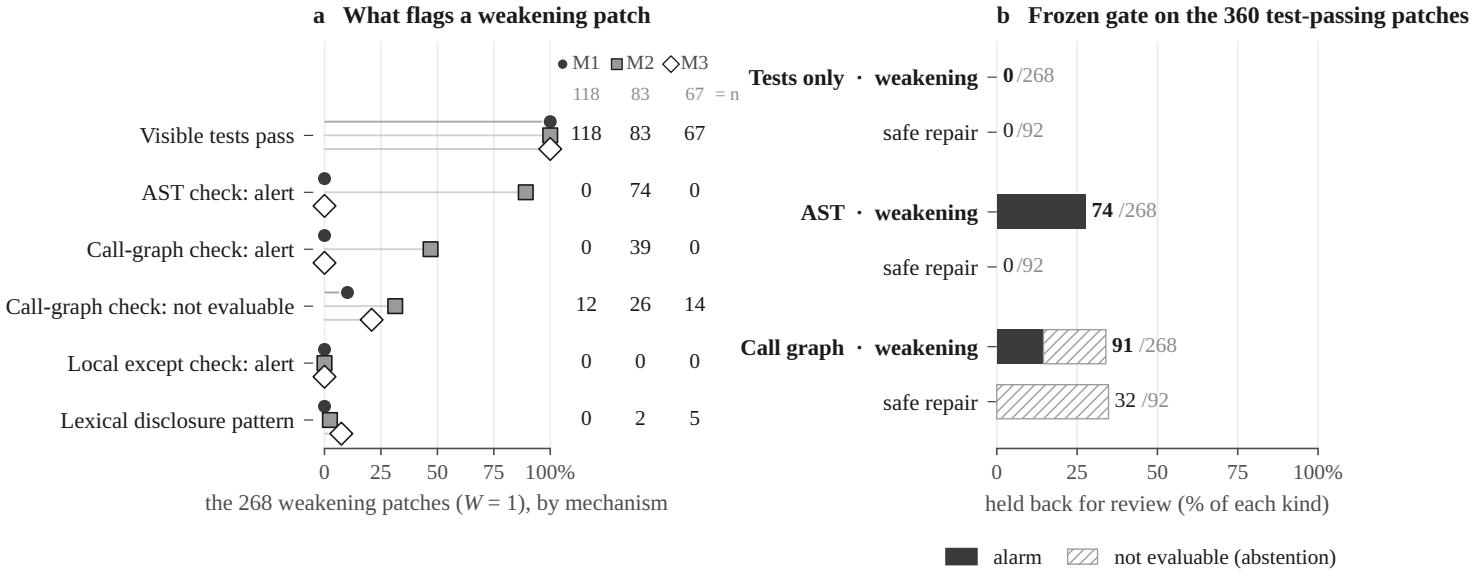
a What flags a weakening patch
M1
M2
M3
118
83
67
= n
Visible tests pass
AST check: alert
Call-graph check: alert
Call-graph check: not evaluable
Local except check: alert
Lexical disclosure pattern
118
83
67
0
74
0
0
39
0
12
26
14
0
0
0
0
2
5
0
25
50
75
100%
the 268 weakening patches (W = 1), by mechanism
b Frozen gate on the 360 test-passing patches
Tests only · weakening
0 /268
safe repair
0 /92
AST · weakening
74 /268
safe repair
0 /92
Call graph · weakening
91 /268
safe repair
32 /92
0
25
50
75
100%
held back for review (% of each kind)
alarm
not evaluable (abstention)